Package 'BOAT'

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Type Package

Title Bayesian Umbrella Trial Design Using Patient-Derived Organoid Data

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Description Implements the BOAT design for randomized umbrella trials in which each patient's tumor gives patient-derived organoids (PDOs) that are tested with every treatment available in the patient's subgroup. Patient responses and PDO responses are modelled jointly, the response mean of each treatment is adaptively clustered across subgroups by reversible jump MCMC, and patients are adaptively randomized according to how well their own PDO responses predict clinical response. The package provides the decision boundaries, the model fit and the interim, final and randomization decisions needed to conduct a trial, simulation of operating characteristics, and calibration of the tuning parameter.

License GPL (>= 3)

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Author Yong Zang [aut], Saijun Zhao [aut, cre]

Maintainer Saijun Zhao <zsj14000639@gmail.com>

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`boat_boundary` *Decision cutoffs of the interim and final analyses*

Description

At an analysis with n_k patients in subgroup B_k, treatment A_j is dropped from B_k if

$$\Pr(\pi_{R,j,k} > \pi_{R,0,k} + \delta_R \mid \mathcal{D}) < c_k \quad \text{or} \quad \Pr(\pi_{T,j} < \pi_{T,0} + \delta_T \mid \mathcal{D}) < c_k, \qquad c_k = \lambda\sqrt{n_k/N_k}.$$

At the final analysis ($n_k = N_k$, so $c_k = \lambda$) a treatment still open is recommended if both probabilities are at least c_k.

Usage

```
boat_boundary(design)
```

Arguments

`design`	A 'boat_design' object.

Value

A data frame with one row per analysis and subgroup: the number of patients and the cutoff.

Examples

```
design <- boat_design(arms = list(c(0, 1, 3), 0:3, c(0, 2, 3)), N = 150, lambda = 0.70)
boat_boundary(design)
```

`boat_calibrate` *Calibrate the tuning parameter lambda*

Description

For each value of 'lambda', simulates the scenarios and computes the family-wise error rate FWER_k of every subgroup without admissible treatment. Larger 'lambda' gives stricter cutoffs and a lower FWER. The chosen λ is the value whose largest FWER is closest to 'alpha'. Ties go to the smallest value with FWER at most 'alpha'.

Usage

```
boat_calibrate(
  design,
  scenarios,
  lambda = seq(0.6, 0.8, by = 0.02),
  nsim = 1000,
  seeds = 1000 + seq_len(nsim),
  alpha = 0.1,
  dgm = boat_dgm(),
  cores = 1
)
```

Arguments

design	A ‘boat_design‘ object.
scenarios	List of ‘boat_scenario‘ objects with at least one null subgroup.
lambda	Values of λ to evaluate.
nsim	Number of simulated trials.
seeds	Seeds of the trials.
alpha	Nominal FWER.
dgm	Data-generating parameters from [boat_dgm()].
cores	Number of cores (forked processes; 1 on Windows).

Value

An object of class ‘boat_calibration‘ with the FWER of every null subgroup for every λ (‘table‘), the largest FWER per λ (‘max_fwer‘) and the chosen value (‘lambda‘).

Examples

```
## Not run:
design <- boat_design(arms = list(c(0, 1, 3), 0:3, c(0, 2, 3)))
# subgroup 2 has no admissible treatment
truth <- boat_scenario(pR = rbind(c(0.50, 0.30, NA, 0.70),
                                  c(0.50, 0.70, 0.50, 0.30),
                                  c(0.30,   NA, 0.50, 0.30)),
                       piT = c(0.25, 0.35, 0.20, 0.15))
boat_calibrate(design, list(truth), lambda = seq(0.64, 0.76, by = 0.02), nsim = 1000, cores = 8)

## End(Not run)
```

---

boat_decide *Interim and final decisions*

Description

Applies the decision rule of [boat_boundary()] to a model fit. At an interim analysis a treatment is dropped from a subgroup if its posterior probability of being effective or of being safe is below the cutoff, and a subgroup with no experimental treatment left stops accrual. At the final analysis the treatments still open are recommended if both probabilities reach the cutoff.

Usage

```
boat_decide(design, fit, n, active = NULL, final = FALSE)
```

Arguments

design	A ‘boat_design‘ object.
fit	A ‘boat_fit‘ object from [boat_fit()].
n	Current number of patients in each subgroup (vector of length ‘K‘).
active	List of the treatments still open in each subgroup (control included). ‘NULL‘ means all treatments of the design.
final	‘TRUE‘ at the final analysis.

Value

An object of class ‘boat_decision‘ with elements ‘table‘ (one row per open experimental treatment with the two probabilities, the cutoff and the decision), ‘active‘ (open treatments after the analysis), ‘stopped‘ (subgroups that stopped accrual) and, at the final analysis, ‘recommended‘ (list of recommended treatments per subgroup).

Examples

```
design <- boat_design(arms = list(c(0, 1, 3), 0:3, c(0, 2, 3)), N = 150, lambda = 0.70,
                      mcmc = list(iter = 1000, burn = 500))
truth <- boat_scenario(pR = rbind(c(0.50, 0.30, NA, 0.70),
                                  c(0.50, 0.70, 0.50, 0.30),
                                  c(0.30,   NA, 0.50, 0.30)),
                       piT = c(0.25, 0.35, 0.20, 0.15))
dat <- boat_generate_data(design, truth, n = 75, seed = 1)
fit <- boat_fit(design, dat$pat, dat$pdo, seed = 1)
boat_decide(design, fit, n = c(75, 75, 75))
```

---

boat_design *Specify a BOAT design*

Description

Collects the design parameters of a BOAT trial.

Usage

```
boat_design(
  K = 3,
  J = 3,
  arms = NULL,
  N = 150,
  interim = c(0.5, 0.75),
  lambda = 0.7,
  delta_T = 0.05,
  delta_R = 0.05,
  randomization = c("adaptive", "equal"),
```

```
  gamma = 1,
  trunc = 0.1,
  mcmc = list(iter = 10000, burn = 5000, thin = 1),
  rar_mcmc = list(iter = 2000, burn = 1000, thin = 1),
  prior = list(),
  rar_standardize = TRUE
)
```

Arguments

K	Number of subgroups.
J	Number of experimental treatments. Treatment 0 is the control.
arms	List of length ‘K‘. Element ‘k‘ gives the treatments evaluated in subgroup ‘k‘; it must contain the control 0. ‘NULL‘ evaluates all treatments ‘0:J‘ in every subgroup.
N	Maximum sample size of each subgroup, a single number or a vector of length ‘K‘.
interim	Fractions of ‘N‘ at which interim analyses are done.
lambda	Tuning parameter of the decision cutoffs (see [boat_boundary()]).
delta_T	Toxicity margin δ_T.
delta_R	Minimum clinically meaningful response improvement δ_R.
randomization	‘"adaptive"‘ for PDO-guided response-adaptive randomization or ‘"equal"‘ for equal randomization among the open treatments.
gamma	Power applied to the randomization probabilities.
trunc	Truncation of the randomization probabilities: they are truncated to ‘[trunc, 1 - trunc]‘ and renormalized. Must be below 1 over the number of open treatments.
mcmc	Iterations, burn-in and thinning of the RJMCMC sampler of the response model.
rar_mcmc	Iterations, burn-in and thinning of the Gibbs sampler of the probit models used for randomization.
prior	List of hyperparameters: ‘mu_mean‘ and ‘mu_sd‘ (normal prior of the probit response means), ‘var_a‘ and ‘var_b‘ (inverse-gamma priors of the variances), ‘rho_mean‘ and ‘rho_sd‘ (truncated normal prior of $\rho > 0$), ‘tox_a‘ and ‘tox_b‘ (beta priors of the toxicity probabilities), ‘rar_mean‘ and ‘rar_sd‘ (normal priors of the intercept and slope of the randomization probit models).
rar_standardize	Standardize the PDO responses in the randomization probit models.

Value

An object of class ‘boat_design‘.

See Also

[boat_boundary()]

Examples

```
# three subgroups and three experimental treatments; A2 is not evaluated in B1 and A1 not in B3
design <- boat_design(K = 3, J = 3, arms = list(c(0, 1, 3), 0:3, c(0, 2, 3)), N = 150,
                      interim = c(0.5, 0.75), lambda = 0.70)
design
```

boat_dgm *Data-generating parameters of a simulation*

Description

Data-generating parameters of a simulation

Usage

```
boat_dgm(sigma_E = 1, rho = 0.6, sigma_O = 0.1, sigma_OR = 0.1, M = 20)
```

Arguments

sigma_E	Standard deviation of the patient random effect shared by the patient's clinical response and PDO responses.
rho	Association between the patient random effect and the PDO responses.
sigma_O	Residual standard deviation of the PDO responses.
sigma_OR	Standard deviation linking the PDO response mean to the clinical response mean on the probit scale.
M	Number of PDO replicates per patient and treatment.

Value

A list of class 'boat_dgm'.

Examples

```
boat_dgm(rho = 0.3)
```

boat_fit *Fit the BOAT model to the trial data*

Description

Fits the joint model of the patients' clinical responses and their PDO responses, with adaptive clustering of each treatment's response mean across subgroups by reversible jump MCMC, and the beta-binomial toxicity model. Used at every interim and at the final analysis.

Usage

```
boat_fit(design, pat, pdo, seed = NULL)
```

Arguments

design	A ‘boat_design‘ object.
pat	Patient data: a data frame with one row per patient and columns ‘id‘, ‘k‘ (subgroup, ‘1:K‘), ‘j‘ (assigned treatment, ‘0‘ = control), ‘YT‘ (toxicity, 0/1) and ‘YR‘ (clinical response, 0/1).
pdo	PDO data: a data frame with one row per PDO replicate and columns ‘id‘ (patient), ‘k‘, ‘j‘ (treatment tested) and ‘YO‘ (PDO response). Each patient’s PDOs are tested with the treatments open in the patient’s subgroup at enrollment.
seed	Seed set before the MCMC, or ‘NULL‘ to use the current random number stream.

Value

An object of class ‘boat_fit‘ with elements

‘summary‘ data frame with one row per subgroup and treatment: posterior mean response probability ‘pR‘, $\Pr(\pi_{R,j,k} > \pi_{R,0,k} + \delta_R)$ (‘pr_eff‘), posterior mean toxicity probability ‘pT‘ and $\Pr(\pi_{T,j} < \pi_{T,0} + \delta_T)$ (‘pr_safe‘).

‘pr_eff‘, ‘pR‘ the same quantities as ‘K‘ x ‘(J + 1)‘ matrices.

‘pr_safe‘, ‘pT‘ vectors over the treatments.

‘partition‘ posterior probabilities of the partitions of the subgroups induced by the clustering of each treatment’s response mean (a pattern such as ‘"112"‘ puts B_1 and B_2 in one cluster).

Examples

```
design <- boat_design(arms = list(c(0, 1, 3), 0:3, c(0, 2, 3)), N = 150, lambda = 0.70,
                      mcmc = list(iter = 1000, burn = 500))
truth <- boat_scenario(pR = rbind(c(0.50, 0.30, NA, 0.70),
                                  c(0.50, 0.70, 0.50, 0.30),
                                  c(0.30,   NA, 0.50, 0.30)),
                       piT = c(0.25, 0.35, 0.20, 0.15))
dat <- boat_generate_data(design, truth, n = 20, seed = 1)
fit <- boat_fit(design, dat$pat, dat$pdo, seed = 1)
fit
```

---

boat_generate_data *Generate trial data*

Description

Generates data of ‘n‘ patients per subgroup under a scenario, with equal randomization among the treatments of each subgroup and the data-generating model of [boat_simulate_trial()]. Useful for trying the functions that conduct a trial.

Usage

```
boat_generate_data(design, scenario, n, dgm = boat_dgm(), seed = NULL)
```

Arguments

design	A ‘boat_design‘ object.
scenario	A ‘boat_scenario‘ object whose evaluated treatments match ‘design$arms‘.
n	Number of patients per subgroup.
dgm	Data-generating parameters from [boat_dgm()].
seed	Random seed of the trial.

Value

A list with the patient data ‘pat‘ (columns ‘id‘, ‘k‘, ‘j‘, ‘YT‘, ‘YR‘) and the PDO data ‘pdo‘ (columns ‘id‘, ‘k‘, ‘j‘, ‘YO‘) in the format of [boat_fit()].

Examples

```
design <- boat_design(arms = list(c(0, 1, 3), 0:3, c(0, 2, 3)))
truth <- boat_scenario(pR = rbind(c(0.50, 0.30, NA, 0.70),
                                  c(0.50, 0.70, 0.50, 0.30),
                                  c(0.30,   NA, 0.50, 0.30)),
                       piT = c(0.25, 0.35, 0.20, 0.15))
dat <- boat_generate_data(design, truth, n = 10, seed = 1)
head(dat$pat)
```

---

boat_get_oc *Operating characteristics*

Description

Simulates ‘nsim‘ trials with [boat_simulate_trial()] and summarizes them by subgroup and treatment.

Usage

```
boat_get_oc(
  design,
  scenario,
  nsim = 1000,
  seeds = 1000 + seq_len(nsim),
  dgm = boat_dgm(),
  cores = 1
)
```

Arguments

design	A ‘boat_design‘ object.
scenario	A ‘boat_scenario‘ object whose evaluated treatments match ‘design$arms‘.
nsim	Number of simulated trials.
seeds	Seeds of the trials.
dgm	Data-generating parameters from [boat_dgm()].
cores	Number of cores (forked processes; 1 on Windows).

Details

For each subgroup and treatment the summary gives the percentage of trials in which the treatment is recommended (‘final_rec‘; for the control, the percentage of trials with no experimental treatment recommended), the mean number of patients (‘n_patients‘) and their percentage (‘pct_patients‘). For a subgroup with admissible treatments it gives the power ‘power1‘ ($P_{k,1}$: some treatment is recommended and all recommended treatments are admissible) and ‘power2‘ ($P_{k,2}$: at least one admissible treatment is recommended). For a subgroup without admissible treatment it gives the family-wise error rate ‘fwer‘.

Value

An object of class ‘boat_oc‘ with the summary ‘table‘ (long format) and the simulated trials ‘trials‘.

Examples

```
fast <- boat_design(arms = list(c(0, 1, 3), 0:3, c(0, 2, 3)), N = 30, lambda = 0.70,
              mcmc = list(iter = 400, burn = 200), rar_mcmc = list(iter = 200, burn = 100))
truth <- boat_scenario(pR = rbind(c(0.50, 0.30, NA, 0.70),
                                  c(0.50, 0.70, 0.50, 0.30),
                                  c(0.30,   NA, 0.50, 0.30)),
                       piT = c(0.25, 0.35, 0.20, 0.15))
boat_get_oc(fast, truth, nsim = 2)
```

---

boat_randomize *Randomize a new patient*

Description

Gives the randomization probabilities of a newly enrolled patient in subgroup ‘k‘ and draws the patient's treatment. Under adaptive randomization, the probability of treatment A_j is proportional to the posterior probability that A_j gives the highest response probability for this patient, predicted from the patient's own mean PDO response to each open treatment. The probabilities are then truncated to ‘[trunc, 1 - trunc]‘ and renormalized (so they can end slightly below ‘trunc‘). Before the first interim analysis (‘rar = NULL‘) and under equal randomization all open treatments have the same probability.

Usage

```
boat_randomize(design, k, pdo_mean = NULL, rar = NULL, active = NULL)
```

Arguments

design	A ‘boat_design‘ object.
k	Subgroup of the new patient.
pdo_mean	The patient's mean PDO response to each treatment open in subgroup ‘k‘: a vector named by treatment (for example ‘c("0" = 0.1, "1" = 0.4, "3" = 0.2)‘). Not needed before the first interim analysis or under equal randomization.
rar	A ‘boat_rar‘ object from [boat_rar_model()], or ‘NULL‘ before the first interim analysis.
active	List of the treatments open in each subgroup. Defaults to ‘rar$active‘ or to all treatments of the design.

Value

A list with the randomization probabilities ‘prob‘ (named by treatment) and the assigned treatment ‘treatment‘.

Examples

```
design <- boat_design(arms = list(c(0, 1, 3), 0:3, c(0, 2, 3)))
# before the first interim analysis all open treatments are equally likely
boat_randomize(design, k = 2)
```

`boat_rar_model` *Fit the randomization model*

Description

For each open experimental treatment and the control in each active subgroup, fits a probit regression of the patients' clinical responses on their mean PDO responses to that treatment (Albert-Chib Gibbs sampler). [boat_randomize()] uses these fits to randomize the patients enrolled until the next analysis. Only used by the adaptive design.

Usage

```
boat_rar_model(design, pat, pdo, active = NULL, seed = NULL)
```

Arguments

`design`	A ‘boat_design‘ object.
`pat`	Patient data: a data frame with one row per patient and columns ‘id‘, ‘k‘ (subgroup, ‘1:K‘), ‘j‘ (assigned treatment, ‘0‘ = control), ‘YT‘ (toxicity, 0/1) and ‘YR‘ (clinical response, 0/1).
`pdo`	PDO data: a data frame with one row per PDO replicate and columns ‘id‘ (patient), ‘k‘, ‘j‘ (treatment tested) and ‘YO‘ (PDO response). Each patient's PDOs are tested with the treatments open in the patient's subgroup at enrollment.
`active`	List of the treatments still open in each subgroup, usually the ‘active‘ element of [boat_decide()]. ‘NULL‘ means all treatments of the design.
`seed`	Seed set before sampling, or ‘NULL‘ to use the current random number stream.

Value

An object of class ‘boat_rar‘.

Examples

```
design <- boat_design(arms = list(c(0, 1, 3), 0:3, c(0, 2, 3)), N = 150, lambda = 0.70,
                      rar_mcmc = list(iter = 400, burn = 200))
truth <- boat_scenario(pR = rbind(c(0.50, 0.30, NA, 0.70),
                                  c(0.50, 0.70, 0.50, 0.30),
                                  c(0.30,   NA, 0.50, 0.30)),
                       piT = c(0.25, 0.35, 0.20, 0.15))
dat <- boat_generate_data(design, truth, n = 30, seed = 1)
rar <- boat_rar_model(design, dat$pat, dat$pdo, seed = 2)
```

```
# a new patient in subgroup 1 whose PDOs responded best to A3
boat_randomize(design, k = 1, pdo_mean = c("0" = -0.2, "1" = 0.1, "3" = 0.8), rar = rar)
```

`boat_scenario` *Specify a simulation scenario*

Description

Specify a simulation scenario

Usage

```
boat_scenario(pR, piT)
```

Arguments

`pR`	Matrix of true response probabilities with one row per subgroup and one column per treatment (first column = control). 'NA' marks a treatment not evaluated in a subgroup.
`piT`	Vector of true toxicity probabilities of the treatments (first = control).

Value

An object of class 'boat_scenario'.

Examples

```
boat_scenario(pR = rbind(c(0.5, 0.3, NA, 0.7), c(0.5, 0.7, 0.5, 0.3), c(0.3, NA, 0.5, 0.3)),
              piT = c(0.25, 0.35, 0.20, 0.15))
```

`boat_simulate_trial` *Simulate one BOAT trial*

Description

Simulates a complete trial under a scenario: patients enter the subgroups in a balanced way, each patient's PDOs are tested with every open treatment of the subgroup, the patient is randomized ([boat_randomize()]), interim analyses ([boat_fit()], [boat_decide()], [boat_rar_model()]) are done when every active subgroup reaches its interim sample size, and the final analysis gives the recommended treatments.

Usage

```
boat_simulate_trial(
  design,
  scenario,
  dgm = boat_dgm(),
  seed = 1001,
  keep_data = FALSE
)
```

Arguments

design	A 'boat_design' object.
scenario	A 'boat_scenario' object whose evaluated treatments match 'design$arms'.
dgm	Data-generating parameters from [boat_dgm()].
seed	Random seed of the trial.
keep_data	Keep the simulated patient and PDO data in the result.

Details

Data are generated from the BOAT model. The patient random effect is $b_i \sim N(0, \sigma_E^2)$, the clinical response is $Y_{R,i} = I(\mu_{R,j,k} + b_i + e_i > 0)$ with $e_i \sim N(0,1)$, and the PDO replicates are $Y_O \sim N(\mu_{O,j,k} + \rho b_i, \sigma_O^2)$ with $\mu_{O,j,k} \sim N(\mu_{R,j,k}, \sigma_{OR}^2)$. The response means $\mu_{R,j,k}$ are set so that the response probabilities equal those of the scenario.

Value

An object of class 'boat_trial' with elements 'final_selection' (recommended treatments by subgroup), 'n_by_trt' (patients per treatment), 'n_by_trt_by_k' (patients per subgroup and treatment), 'n_by_k', 'stopped_groups', 'interims' (decision tables of the interim analyses) and, if 'keep_data = TRUE', 'pat' and 'pdo'.

Examples

```
fast <- boat_design(arms = list(c(0, 1, 3), 0:3, c(0, 2, 3)), N = 30, lambda = 0.70,
               mcmc = list(iter = 400, burn = 200), rar_mcmc = list(iter = 200, burn = 100))
truth <- boat_scenario(pR = rbind(c(0.50, 0.30, NA, 0.70),
                                  c(0.50, 0.70, 0.50, 0.30),
                                  c(0.30,   NA, 0.50, 0.30)),
                       piT = c(0.25, 0.35, 0.20, 0.15))
boat_simulate_trial(fast, truth, seed = 1001)
```

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